Assignment 3 – Antonin Beranger

Income distribution (ECON 473)

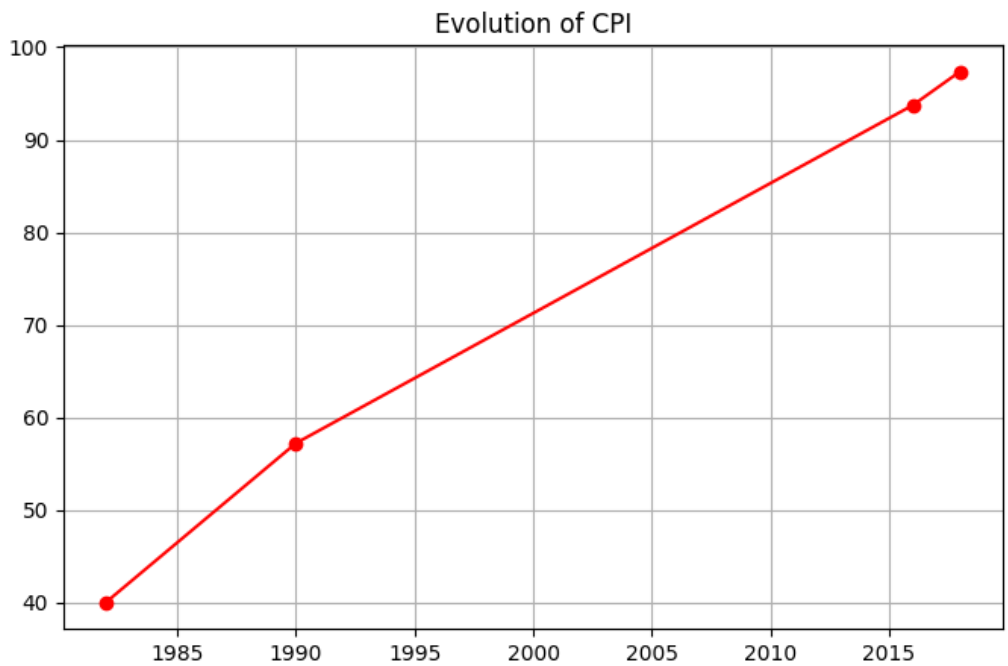
Question 1:

1982 – 1990:	2016-2018:
*Total Income: TOTINC	*Total Income: TTINC
*After-tax income: INCAFTTX	*After-tax income: ATINC
*Wages and Salaries: WAGSAL	*Wages and Salaries: WGSAL
*Hours worked: HRSWRK	*Hours worked: USHRWK
*Weeks worked: WKSWRK	*Weeks worked: WKSEM
*Education: EDUC	*Education: HLEV2G
*Age: AGE	*Age: AGEGP
*Gender: SEX	*Gender: SEX
*WEIGHTS: WEIGHT	*WEIGHTS: FWEIGHT

Hourly wage:

➔ `df['hourly_wage'] = df[wage_var] / (df[hours_var] * df[weeks_var])`

Question 2:




```
Real hourly wage in 2020 $:
- cpi = {1982: 40.07, 1990: 57.23, 2016: 93.72, 2018: 97.37}
- for y, df in datasets.items():
- df['hourly_real_wage'] = (df['hourly_wage'] / cpi[y]) * 100
```

Question 3:

```
#question 3 - sample selection and replication of table 1
unfiltered_datasets = {year: df.copy() for year, df in datasets.items()}
```

```python
min_wage_thresholds = [3.7375*0.5, 5.0500001*0.5, 11.3*0.5, 13.4125*0.5]
years = [1982, 1990, 2016, 2018]

for year, wage in zip(years, min_wage_thresholds):
    df = datasets[year]

    print("\n" + "="*65)
    print(f"Table 1: \n Sample Selection in the {year} Dataset")
    print("="*65)

    if year in [1982, 1990]:
        hours_var, weeks_var, age_var = 'HRSWRK', 'WKSWRK', 'AGE'
    else:
        hours_var, weeks_var, age_var = 'USHRWK', 'WKSEM', 'AGEGP'

    n_total = len(df)

    # Wage & Work Filter
    df_clean = df[
        (df['hourly_wage'].notna()) &
        (df['hourly_wage'] >= wage) &
        ((df[hours_var] * df[weeks_var]) > 260)
    ]
    n_after_wage = len(df_clean)
    drop1 = n_total - n_after_wage

    # Age Filter
    if year in [1982, 1990]:
        df_clean = df_clean[(df_clean[age_var] >= 25) & (df_clean[age_var] <= 60)]
    else:
        df_clean = df_clean[(df_clean[age_var] >= 7) & (df_clean[age_var] <= 14)]

    n_after_age = len(df_clean)
    drop2 = n_after_wage - n_after_age

    # table 1 replication
    summary = pd.DataFrame({
        'Step': ['Initial data', 'After wage filtering', 'After age filtering'],
        'Observations': [n_total, n_after_wage, n_after_age],
        'Dropped': [0, drop1, drop2],
        'Cumulative Dropped': [0, drop1, drop1 + drop2]
    })
    print(summary.to_string(index=False))
    print("="*65)

    datasets[year] = df_clean
```

Table 1 reproduction from HPV

```
=====================================================================
Table 1:
 Sample Selection in the 1982 Dataset
=====================================================================
                 Step  Observations  Dropped  Cumulative Dropped
         Initial data         42570        0                   0
After wage filtering         34256     8314                8314
 After age filtering         25940     8316               16630
=====================================================================

=====================================================================
Table 1:
 Sample Selection in the 1990 Dataset
=====================================================================
                 Step  Observations  Dropped  Cumulative Dropped
         Initial data         46510        0                   0
After wage filtering         38145     8365                8365
 After age filtering         31350     6795               15160
=====================================================================

=====================================================================
Table 1:
 Sample Selection in the 2016 Dataset
=====================================================================
                 Step  Observations  Dropped  Cumulative Dropped
         Initial data         34287        0                   0
After wage filtering         28327     5960                5960
 After age filtering         23517     4810               10770
=====================================================================

=====================================================================
Table 1:
 Sample Selection in the 2018 Dataset
=====================================================================
                 Step  Observations  Dropped  Cumulative Dropped
         Initial data         50588        0                   0
After wage filtering         41807     8781                8781
 After age filtering         32469     9338               18119
=====================================================================
```

Question 4 and 5 :

Using information from the dataset of 1982: EDUC variable

1	NO SCHOOLING OR ELEMENTARY
2	9 OR 10 YEARS OF ELEMENTARY AND SECONDARY
3	11 YEARS OF ELEMENTARY AND SECONDARY
4	12 YEARS OF ELEMENTARY AND SECONDARY
5	13 YEARS OF ELEMENTARY AND SECONDARY
6	SOME POST-SECONDARY
7	POST-SECONDARY CERTIFICATE OR DIPLOMA
8	UNIVERSITY DEGREE

Using information from the dataset of 1990 : RECEDUC variable

1	NO SCHOOLING OR GRADE 8 OR LOWER,NO OTH EDUC
2	GRADE 9-10, NO OTHER EDUCATION
3	GRADE 11-13,DID NOT GRADUATE,NO OTHER EDUC
4	GRADE 11-13,GRADUATED,NO OTHER EDUCATION
5	SOME POST-SECONDARY,NO DEGREE,CERT,DIPL
6	POST-SECONDARY CERT,DIPL(INCLUDED TRADES)
7	UNIVERSITY DEGREE

Using information from the dataset of 2016 and 2018: HLEV2G variable

1	Less than high school graduation
2	Graduated high school or partial postsecondary education
3	Non-university postsecondary certificate or diploma
4	University degree or certificate
6+	Skip

```
for year, df in datasets.items():
    print(f"Generating Education and Experience for {year}...")

    if year == 1982:
        edu_map = {1: 10, 2: 10, 3: 10, 4: 12, 5: 12, 6: 12, 7: 13, 8: 16}
        df['yrs_schooling'] = df['EDUC'].map(edu_map)
        df['pot_exp'] = df['AGE'] - np.maximum(10, df['yrs_schooling']) - 6

    elif year == 1990:
        edu_map = {1: 10, 2: 10, 3: 10, 4: 12, 5: 12, 6: 13, 7: 16}
        df['yrs_schooling'] = df['RECEDUC'].map(edu_map)
        df['pot_exp'] = df['AGE'] - np.maximum(10, df['yrs_schooling']) - 6

    else:
        edu_map = {1: 10, 2: 12, 3: 13, 4: 16}
        df['yrs_schooling'] = df['HLEV2G'].map(edu_map)
        age_group_to_midpoint = {7: 27, 8: 32, 9: 37, 10: 42, 11: 47, 12: 52, 13:
57, 14: 62}
        df['AGE_numeric'] = df['AGEGP'].map(age_group_to_midpoint)
        df['pot_exp'] = df['AGE_numeric'] - np.maximum(10, df['yrs_schooling']) - 6

    df['yrs_schooling'] = df['yrs_schooling'].fillna(10)
```

```
    df['pot_exp'] = df['pot_exp'].clip(lower=0)
    df = df.dropna(subset=['real_wage', 'yrs_schooling', 'pot_exp'])
    datasets[year] = df
```

Question 6:

Between 1982 and 2018, wage inequality steadily increased, as shown by the rising log variance of wages. However, the inequality of its main determinants did not follow this trend: the variance in years of schooling decreased as education levels became more uniform, and the variance in experience ended up roughly where it started. Because the distribution of education and experience did not become more unequal over time, the evolution of these specific determinants cannot explain the widening wage gap.

```
Question 6 Table
 Year  Mean Real Wage  Log Var Wages  Mean Schooling  Var Schooling  Mean Experience  Var Experience
 1982         24.9541         0.2846         11.8980         4.0415          21.3397        112.0865
 1990         26.4512         0.3103         11.9349         0.1260          20.9397         88.7488
 2016         33.6662         0.3562         13.6578         3.6334          23.4534        131.5907
 2018         36.2817         0.3498         13.7230         3.5357          21.8667        114.5835
```

Question 7:

```
def run_mincer_regression(df):
    """
    This helper fits a Mincer regression and returns the results object.
    Expects df to have: log_real_wage, yrs_schooling, pot_exp, final_weight
    """
    df = df.copy()
    df['pot_exp_sq'] = df['pot_exp'] ** 2

    Y = df['log_real_wage']
    X = sm.add_constant(df[['yrs_schooling', 'pot_exp', 'pot_exp_sq']])
    w = df['final_weight']

    model = sm.WLS(Y, X, weights=w, missing='drop')
    results = model.fit()

    return results
```

A) Regression Results :

Below are the results of the basic Mincer regression for each sample year. The constant ($ß_0$) is the intercept, then the 3 coefficients ($ß_1$, $ß_2$, $ß_3$) respectively for education, experience and experience squared. Then the R^2 and the number of observations.

```
Question 7a Regression Results

Year: 1982
Constant: 1.8162
Years of Schooling: 0.0840
Potential Exp: 0.0218
Potential Exp^2: -0.0004
R-squared: 0.0858
Observations: 25940.0000

Year: 1990
Constant: 1.5785
Years of Schooling: 0.0941
Potential Exp: 0.0290
Potential Exp^2: -0.0004
R-squared: 0.1024
Observations: 31350.0000

Year: 2016
Constant: 1.4908
Years of Schooling: 0.1015
Potential Exp: 0.0343
Potential Exp^2: -0.0005
R-squared: 0.1162
Observations: 23365.0000

Year: 2018
Constant: 1.7476
Years of Schooling: 0.0995
Potential Exp: 0.0253
Potential Exp^2: -0.0005
R-squared: 0.0973
Observations: 31317.0000
```

B) Marginal Return:

```
results = regression_results[2018]
mean_experience = final_table.loc[final_table['Year'] == 2018, 'Mean
Experience'].values[0]
marginal_return = results['Potential Exp'] + 2 * results['Potential Exp^2'] *
mean_experience
```

Marginal return to experience in 2018 at mean experience level (21.87 years) is 0.0055

C) Findings:

The returns to both schooling and experience increased over time, with the wage premium for an additional year of education rising from 8.4% in 1982 to roughly 9.95% in 2018. The initial return to a year of experience also grew from 2.18% to 2.53% over the same period.

The negative coefficient on the experience squared term indicates diminishing returns to experience, but the overall increase in the linear term suggests that the early years of experience became more valuable in the labor market.

Yes, these rising returns help explain the evolution of wage inequality:
Even dough the workforce's education level became more equal over time, the labor market began paying a much higher premium for the remaining differences in human capital, which directly widens the overall wage gap. Despite these rising returns, the low R^2 values indicate that schooling and experience explain very little of overall wage inequality, pointing to unobserved factors as the main driver of wage dispersion in Canada.

Question 8:

```
# 1st regression : Base Mincer + Female Control
print("\nRegression 1: Base Mincer + Female Control\n")

spec1_results = {}

for year in [1982, 1990, 2016, 2018]:
    df = datasets[year]
    Y = df['log_real_wage']
    X1 = sm.add_constant(df[['yrs_schooling', 'pot_exp', 'pot_exp_sq', 'female']])
    w = df['final_weight']

    model1 = sm.WLS(Y, X1, weights=w, missing='drop').fit()
    spec1_results[year] = {
        'Constant': model1.params['const'],
        'Years of Schooling': model1.params['yrs_schooling'],
        'Potential Exp': model1.params['pot_exp'],
        'Potential Exp^2': model1.params['pot_exp_sq'],
        'Female Dummy': model1.params['female'],
        'R-squared': model1.rsquared,
        'Observations': int(model1.nobs)
    }

spec1_table = pd.DataFrame(spec1_results)
pd.options.display.float_format = '{:,.4f}'.format
print(spec1_table.to_string())

# 2nd regression : Base Mincer + Female Control + Interaction Term
print("\nRegression 2: Base Mincer + Female Control + (Education x Experience)\n")

spec2_results = {}

for year in [1982, 1990, 2016, 2018]:
    df = datasets[year]
    Y = df['log_real_wage']
    X2 = sm.add_constant(df[['yrs_schooling', 'pot_exp', 'pot_exp_sq', 'female',
'edu_exp']])
    w = df['final_weight']
```

```
    model2 = sm.WLS(Y, X2, weights=w, missing='drop').fit()
    spec2_results[year] = {
        'Constant': model2.params['const'],
        'Years of Schooling': model2.params['yrs_schooling'],
        'Potential Exp': model2.params['pot_exp'],
        'Potential Exp^2': model2.params['pot_exp_sq'],
        'Female Dummy': model2.params['female'],
        'Edu x Exp': model2.params['edu_exp'],
        'R-squared': model2.rsquared,
        'Observations': int(model2.nobs)
    }

spec2_table = pd.DataFrame(spec2_results)
print(spec2_table.to_string())
```

```
Regression 1: Base Mincer + Female Control

                         1982        1990        2016        2018
Constant               1.9458      1.7167      1.5240      1.7690
Years of Schooling     0.0849      0.0937      0.1073      0.1053
Potential Exp          0.0217      0.0292      0.0335      0.0254
Potential Exp^2       -0.0004     -0.0004     -0.0005     -0.0005
Female Dummy          -0.3489     -0.2954     -0.2258     -0.2227
R-squared              0.1870      0.1717      0.1515      0.1323
Observations       25,940.0000 31,350.0000 23,365.0000 31,317.0000

Regression 2: Base Mincer + Female Control + (Education x Experience)

                         1982        1990        2016        2018
Constant               2.1029      1.6412      1.5173      1.6418
Years of Schooling     0.0738      0.0990      0.1077      0.1135
Potential Exp          0.0124      0.0335      0.0338      0.0320
Potential Exp^2       -0.0003     -0.0005     -0.0005     -0.0005
Female Dummy          -0.3491     -0.2957     -0.2258     -0.2237
Edu x Exp              0.0006     -0.0003     -0.0000     -0.0004
R-squared              0.1874      0.1717      0.1515      0.1325
Observations       25,940.0000 31,350.0000 23,365.0000 31,317.0000
```

1st regression: The basic Mincer model assumes that the returns to education and experience are the same for everyone. However, labor economics literature shows a persistent gender wage gap. Including the female dummy allows the intercept to shift, capturing the average difference in log wages between men and women that isn't explained by schooling or experience.

2nd regression: I added the interaction term edu_exp to test the complementarity of human capital. The logic is that education might make someone better at learning on the job, meaning an extra year of experience is more valuable for a university graduate than for a high school dropout. This changes the functional form from a simple additive model to one where the return to schooling depends on how long you've been in the workforce.

In my opinion, while the interaction term in regression 2 is theoretically interesting, it often introduces significant multicollinearity between education, experience and the interaction term,

which can make the individual coefficients for schooling and experience unstable. Adding the female dummy in regression 1 significantly improves the model's fit and provides a direct, easy-to-interpret measure of the gender pay gap over time. This allows us to see the gender gap decreasing from 1982 to 2018, probably because of women achieving higher skill levels and getting better paid jobs.

Question 9:

$$\log(\text{wage}) = \beta_0 + \beta_1 * \text{schooling} + \beta_2 * \text{experience} + \beta_3 * \text{experience}^2 + \text{epsilon}$$

In the Mincer regression, the log of wages is modeled as a function of schooling and experience (both linear and squared), with an error term (epsilon) capturing unobserved factors.

1. Changes in Attributes: If the average level of schooling or experience in the population changes, this will affect the average log wage. For example, if more people attain higher education, the average log wage would increase due to a higher β_1*schooling term.

2. Changes in Returns: If the coefficients (β_1, β_2, β_3) change over time, this indicates that the labor market is valuing these attributes differently. For instance, if the return to schooling (β_1) increases, even if the average schooling level remains the same, the average log wage would increase because each year of schooling is now worth more in terms of wages.

3. Changes in Residuals: The error term captures unobserved factors that affect wages. If the distribution of these unobserved factors changes over time, this could also lead to changes in the average log wage, independent of changes in attributes or returns.

Question 10:

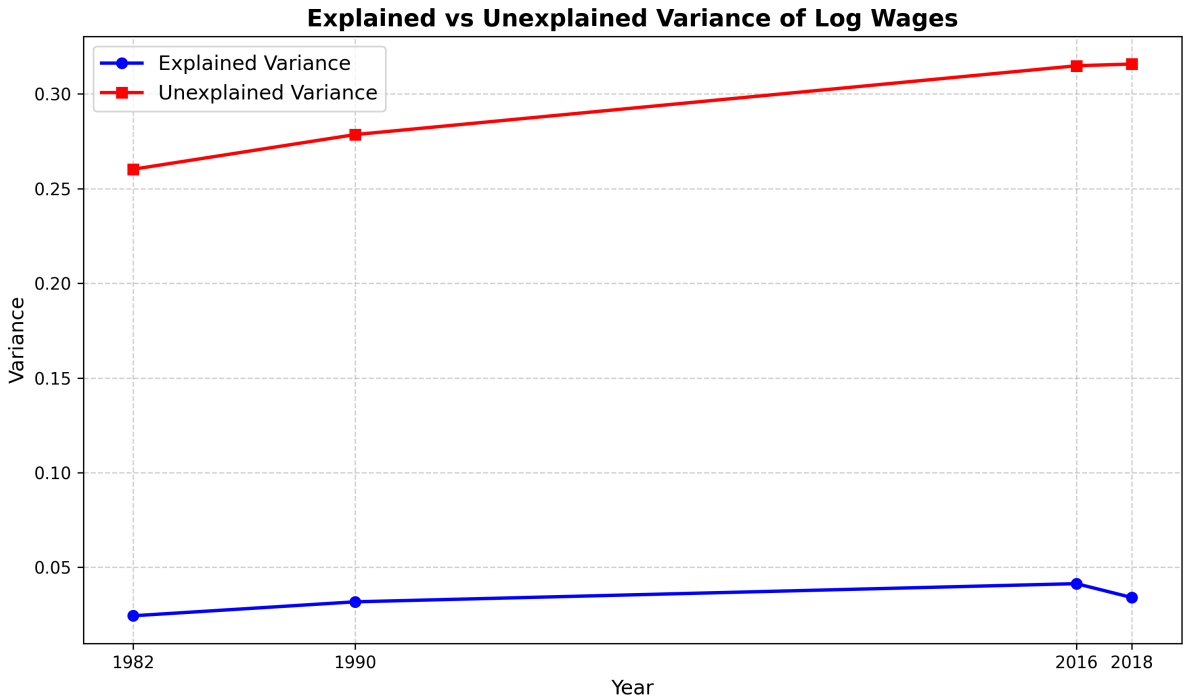
```
R-squared for Base Mincer Regression:

1982: 0.0858
1990: 0.0236
2016: 0.1162
2018: 0.0973
```

```
Explanatory power of the regression:

1982: Total Var = 0.2846 | Explained = 0.0244 | Unexplained = 0.2602
1990: Total Var = 0.3103 | Explained = 0.0318 | Unexplained = 0.2785
2016: Total Var = 0.3562 | Explained = 0.0414 | Unexplained = 0.3148
2018: Total Var = 0.3498 | Explained = 0.0340 | Unexplained = 0.3158
```

```
exp_var = total_log_var * r_squared
unexp_var = total_log_var * (1 - r_squared)
```



As overall variance grew from 0.285 to 0.35 between 1982 and 2018, the unexplained variance increased much more than the explained variance. Between 1982 and 2018, the explained inequality grew by roughly 0.0096 (from 0.0244 to 0.0340), while the unexplained inequality grew by 0.0556 (from 0.2602 to 0.3158). This means that while education and experience play a role, most of the rise in Canadian wage inequality over this period was driven by unobserved factors (technological change, decline in unionization, or shifts in industry premiums) that are not captured in the basic Mincer equation.

Question 11:

```
df_1982 = datasets[1982].copy()
df_1982['year_2018'] = 0

df_2018 = datasets[2018].copy()
df_2018['year_2018'] = 1

combined_df = pd.concat([df_1982, df_2018], ignore_index=True)

combined_df['school_x_2018'] = combined_df['yrs_schooling'] *
combined_df['year_2018']

Y = combined_df['log_real_wage']
X = sm.add_constant(combined_df[['yrs_schooling', 'pot_exp', 'pot_exp_sq',
'year_2018', 'school_x_2018']])
w = combined_df['final_weight']
```

	coef	std err	t	P>|t|	[0.025	0.975]
const	1.8138	0.296	6.118	0.000	1.233	2.395
yrs_schooling	0.0825	0.025	3.360	0.001	0.034	0.131
pot_exp	0.0253	0.001	25.695	0.000	0.023	0.027
pot_exp_sq	-0.0005	2.11e-05	-21.450	0.000	-0.000	-0.000
year_2018	-0.0663	0.297	-0.223	0.823	-0.648	0.515
school_x_2018	0.0170	0.025	0.691	0.490	-0.031	0.065

This model tests whether the return to schooling changed in 2018.

From the table, the coefficient for school_x_2018 is 0.0170 and the p-value is 0.490.

The coefficient is not statistically significant, suggesting there is no evidence that the effect of schooling on log(wage) changed in 2018. However, the estimated coefficient of 0.0170 suggests that in 2018 the marginal effect of an additional year of schooling was 0.017 higher than in the base year. Since the coefficient on schooling is 0.0825, the total return to an additional year of schooling in 2018 is 0.0825 + 0.0170 = 0.0995. This adds to the coefficient found in the first Mincer regression for schooling for 2018.

Question 12:

To isolate the contribution of changing returns from changing attributes, we apply the 1982 regression coefficients (prices) to the 2018 distribution of schooling and experience (attributes), generating counterfactual predicted wages that represent what workers in 2018 would have earned if the labor market had still valued their human capital at 1982 rates. The variance of these counterfactual predicted wages then measures how much explained inequality the 2018 workforce would have produced under 1982 prices alone. If this number is close to the actual 1982 explained inequality, it means the change in the attribute distribution contributed little, and that rising returns are what drove the increase in explained inequality.

```
beta_1982 = [
    regression_results[1982]['Constant'],
    regression_results[1982]['Years of Schooling'],
    regression_results[1982]['Potential Exp'],
    regression_results[1982]['Potential Exp^2']
]

df_2018 = datasets[2018].copy()
df_2018['pot_exp_sq'] = df_2018['pot_exp'] ** 2

X_2018 = sm.add_constant(df_2018[['yrs_schooling', 'pot_exp', 'pot_exp_sq']])

predicted_log_wages_1982_prices = np.dot(X_2018, beta_1982)

explained_ineq_quest_12 = weighted_variance(
    predicted_log_wages_1982_prices,
    df_2018['final_weight']
)

print("\nQuestion 12 Analysis:\n")
print(f"Explained Inequality in 1982 (Actual): {explained_vars[0]:.4f}")
```

```
print(f"Explained Inequality in 2018 (Actual): {explained_vars[3]:.4f}")
print(f"Explained Inequality (2018 attributes at 1982 prices):
{explained_ineq_quest_12:.4f}")
```

```
Question 12 Analysis:

Explained Inequality in 1982 (Actual): 0.0244
Explained Inequality in 2018 (Actual): 0.0340
Explained Inequality (2018 attributes at 1982 prices): 0.0240
```

The explained inequality in 2018 using 1982 returns is 0.0240, which is very close to the actual explained inequality in 1982 (0.0244). This suggests that the changes in the distribution of attributes (schooling and experience) between 1982 and 2018 would have led to a similar level of explained wage inequality if the returns to those attributes had remained constant at their 1982 levels. Therefore, the increase in explained wage inequality from 1982 to 2018 is largely driven by changes in the returns to schooling and experience, rather than changes in the distribution of those attributes. This reinforces the conclusion from question 10 that rising returns to human capital are a key factor in the evolution of wage inequality over this period.

Question 13, 14:

```
residuals_dict = {}
residuals_variance = {}

for year in [1982, 1990, 2016, 2018]:
    results = run_mincer_regression(datasets[year])

    year_residuals = results.resid
    residuals_dict[year] = year_residuals

    weights_used = results.model.weights
    resid_var = weighted_variance(year_residuals, weights_used)

    total_var = final_table.loc[final_table['Year'] == year, 'Log Var
Wages'].values[0]
    fraction = resid_var / total_var

    residuals_variance[year] = {
        'Residual Var': resid_var,
        'Fraction': fraction
    }

print("\nQuestion 13: Residuals from Mincer Regressions\n")
for year in [1982, 1990, 2016, 2018]:
    print(f"Sample residuals for {year}: \n{residuals_dict[year][:5]}")

print("\nQuestion 14: Variance of Residuals and Fraction of Total Inequality\n")
for year in [1982, 1990, 2016, 2018]:
    stats = residuals_variance[year]
    print(f"Year {year}:")
    print(f"  Unexplained Variance: {stats['Residual Var']:.4f}")
```

```
    print(f"  Fraction of Total Variance: {stats['Fraction']:.2%}")
```

```
Question 14: Variance of Residuals and Fraction of Total Inequality

Year 1982:
  Unexplained Variance: 0.2602
  Fraction of Total Variance: 91.42%
Year 1990:
  Unexplained Variance: 0.2785
  Fraction of Total Variance: 89.76%
Year 2016:
  Unexplained Variance: 0.3148
  Fraction of Total Variance: 88.38%
Year 2018:
  Unexplained Variance: 0.3158
  Fraction of Total Variance: 90.27%
```

The variance of the residuals from the Mincer regression increased from 0.2602 in 1982 to 0.3158 in 2018, which is a significant increase. The fraction of total wage variance that is unexplained by the Mincer regression is about 90%, staying constant over time, with a spike in 1990 that is not significant. This may suggest that in the 1990 sample, education and experience had almost no predictive power, or the data for that year was much noisier than the other years.

These high percentages indicate that most of the wage variation in Canada occurs between individuals who possess the same observed levels of education and experience. This statement seems valid for all year, with no clear trend over time. This reinforces the conclusion that while education and experience are important determinants of wages, there are many other unobserved factors (such as industry, occupation, skills, discrimination, or luck) that play a much larger role in driving wage inequality.

Question 15:

These findings reveal a nuanced relationship between the Canadian labor market and the U.S. experience described by Autor, Katz, and Kearney. AKK document a rise in the college wage premium in the United States, even if they note that this was not a uniform or uninterrupted trend, showing that the premium plateaued for college-only workers after 1987. In Canada, the return to schooling increased slightly from 8.25% to 9.95% between 1982 and 2018, though this change was not statistically significant, suggesting that the supply of educated workers in Canada kept pace with demand throughout the period.

Question 12 analysis shows that the rise in explained inequality from 1982 to 2018 is almost entirely driven by changing returns rather than changing attributes, directly paralleling AKK's finding that price changes dominate composition changes. Combined with question 14, reporting 90% residual share, this supports a similar conclusion to AKK: modern wage inequality is largely a within-group phenomenon.

Question 16:

All models were built starting from the 1st regression of question 8, adding a female dummy variable to the basic Mincer Regression.

Regression A: Quadratic Schooling

	1982	1990	2016	2018
Constant	1.6809	2.2506	1.0662	2.1028
Schooling	0.1275	0.0098	0.1759	0.0555
P-val (Schooling)	0.0000	0.5938	0.0000	0.0386
Schooling^2	-0.0017	0.0032	-0.0025	0.0018
P-val (Schooling^2)	0.0271	0.0000	0.0210	0.0632
Exp	0.0216	0.0295	0.0331	0.0257
P-val (Exp)	0.0000	0.0000	0.0000	0.0000
Exp^2	-0.0004	-0.0005	-0.0005	-0.0005
P-val (Exp^2)	0.0000	0.0000	0.0000	0.0000
Female	-0.3504	-0.2941	-0.2262	-0.2225
P-val (Female)	0.0000	0.0000	0.0000	0.0000
R-squared	0.1872	0.1722	0.1517	0.1324
Observations	25,940.0000	31,350.0000	23,365.0000	31,317.0000

Model A, Quadratic Schooling (Weakest) : Adding the squared schooling term provided the no explanatory power. While it was significant for 1982, 1990 and 2016, by 2018 the p-value rose to 0.0632, making it statistically insignificant. Additionally, the R-squared barely improved compared to the base Mincer model from earlier questions, meaning it does not explain much of the residual variance.

Regression B: Marital Status (Marriage Premium)

	1982	1990	2016	2018
Constant	1.9301	1.7121	1.5361	1.7401
Schooling	0.0849	0.0937	0.1049	0.1023
P-val (Schooling)	0.0000	0.0000	0.0000	0.0000
Exp	0.0217	0.0292	0.0300	0.0236
P-val (Exp)	0.0000	0.0000	0.0000	0.0000
Exp^2	-0.0004	-0.0005	-0.0004	-0.0004
P-val (Exp^2)	0.0000	0.0000	0.0000	0.0000
Female	-0.3476	-0.2952	-0.2243	-0.2200
P-val (Female)	0.0000	0.0000	0.0000	0.0000
Married	0.0153	0.0043	0.0921	0.1194
P-val (Married)	0.1852	0.6831	0.0000	0.0000
R-squared	0.1871	0.1717	0.1561	0.1399
Observations	25,940.0000	31,350.0000	23,365.0000	31,317.0000

Model B, Marital Status (Strong) : The marriage premium shows an interesting trend: it has a stronger explanatory power but is statistically significant only in the recent data. In 1982 and 1990, the p-values are high (0.1852 and 0.6831), making marital status statistically insignificant. However, in 2016 and 2018, the p-values drop to 0, and the coefficient jumps to nearly 14 percent. Its VIF of 1.01 confirms it is a perfectly independent predictor with zero collinearity issues.

Regression C: Provinces (Reference: Ontario)

	1982	1990	2016	2018
Constant	1.9850	1.7639	1.5209	1.7820
Schooling	0.0823	0.0936	0.1076	0.1045
P-val (Schooling)	0.0000	0.0000	0.0000	0.0000
Exp	0.0217	0.0298	0.0335	0.0261
P-val (Exp)	0.0000	0.0000	0.0000	0.0000
Exp^2	-0.0004	-0.0005	-0.0005	-0.0005
P-val (Exp^2)	0.0000	0.0000	0.0000	0.0000
Female	-0.3474	-0.2965	-0.2248	-0.2205
P-val (Female)	0.0000	0.0000	0.0000	0.0000
Alberta	0.0509	-0.0835	0.1318	0.1244
P-val (Alberta)	0.0000	0.0000	0.0000	0.0000
BC	0.1139	-0.0790	-0.0016	-0.0075
P-val (BC)	0.0000	0.0000	0.8874	0.4430
Manitoba	-0.0845	-0.1030	-0.0523	-0.0677
P-val (Manitoba)	0.0000	0.0000	0.0084	0.0001
New Brunswick	-0.1014	-0.1486	-0.1092	-0.1305
P-val (New Brunswick)	0.0000	0.0000	0.0000	0.0000
Newfoundland	-0.0627	-0.1692	0.0151	-0.0403
P-val (Newfoundland)	0.0000	0.0000	0.6094	0.1173
Nova Scotia	-0.1418	-0.1656	-0.1093	-0.1239
P-val (Nova Scotia)	0.0000	0.0000	0.0000	0.0000
PEI	-0.2145	-0.2159	-0.1727	-0.1918
P-val (PEI)	0.0000	0.0000	0.0017	0.0001
Quebec	0.0381	-0.0558	-0.0673	-0.0797
P-val (Quebec)	0.0001	0.0000	0.0000	0.0000
Saskatchewan	-0.0430	-0.1540	0.0499	0.0361
P-val (Saskatchewan)	0.0004	0.0000	0.0198	0.0603
R-squared	0.2078	0.1798	0.1630	0.1449
Observations	25,940.0000	31,350.0000	23,365.0000	31,317.0000

Model C, Provinces (Strongest) : The geographic fixed effects show great explanatory power. The p-values for provinces like Alberta and Quebec are 0.0 in every single year, meaning the regional wage differences are highly statistically significant. This model also provides the largest jump in overall R-squared. Furthermore, VIF tests show all province scores are under 1.5, proving that geography provides completely unique information about wage inequality that education and experience cannot explain.

Overall, Model C demonstrates that geographic fixed effects have strong and consistent explanatory power, yielding the highest R-squared values of the three models tested. With p-values at or near zero across all decades, these regional wage differences are highly statistically significant. Compared to the Ontario baseline, the coefficients capture major economic shifts over time: Alberta shows a massive 13.2 to 12.4 percent wage advantage in 2016 and 2018, likely reflecting the provincial resource boom. Conversely, wages in Quebec shifted to a persistent 7 to 8 percent penalty relative to Ontario in the later decades. This confirms that local labor market conditions explain a substantial portion of Canadian wage inequality beyond basic education and experience. We can see the R^2 slowly decreasing from 1982 to 2018, suggesting that this model explained less variance in the log(wage). More predictors should be added such as industry, luck or a different measure of skills. Unfortunately, while 1982 and 1990 surveys have the industry category the 2016 and 2018 don't have it. I ran the regression for the 1982 and 1990 data only.

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Regression D: Female + Province + Industry (1982 & 1990)
Note: *** p<0.01  ** p<0.05  * p<0.10  |  ref: Ontario, Agriculture
Variable                                1982         1990
-------------------------------------------------------
--- Base Controls ---
Schooling                          0.0698***    0.0852***
Female                            -0.2945***   -0.2644***
--- Selected Provinces (ref: Ontario) ---
Alberta                            0.0635***   -0.0809***
British Columbia                   0.1173***   -0.0665***
Quebec                             0.0349***   -0.0525***
Nova Scotia                       -0.1509***   -0.1637***
PEI                               -0.2125***   -0.1967***
--- Selected Industries (ref: Agriculture) ---
Other Primary                      0.5904***    0.5799***
Transport & Comm.                  0.5548***    0.4566***
Manufacturing (Dur.)               0.4969***    0.3972***
Retail Trade                       0.2850***    0.2095***
Personal Services                  0.4255***       0.0152
-------------------------------------------------------
R-squared                             0.2508       0.2236
Observations                          23,516       31,350
```

Adding province and industry controls substantially improves model fit, with R-squared rising to 0.25 in 1982 and 0.22 in 1990 compared to roughly 0.09 in the base Mincer regression. The female wage penalty remains large and significant at around -29% and -26%, confirming a persistent gender gap. The most striking provincial result is the sign reversal for Alberta, BC, and Quebec, which went from wage premiums over Ontario in 1982 to penalties in 1990, likely reflecting the collapse of oil prices and the broader recession hitting resource-dependent western provinces. Atlantic provinces like PEI and Nova Scotia show a consistent and significant wage penalty throughout. On the industry side, Other Primary and Transport consistently command the highest premiums above agriculture, while the most notable finding is Personal Services, which had a significant 43% premium in 1982 but became statistically indistinguishable from agriculture by 1990, suggesting this sector was hit particularly hard over the decade. Overall, geography and industry affiliation explain a meaningful share of wage inequality beyond education and experience alone.